component of spin-down, and vice versa. Even the wave function for the position of a particle is a vector, though we normally visualize it as a smooth function throughout space. The trick is to think of every point in space as defining a different component, and the wave function is a superposition of all of them. There are an infinite number of such vectors, so the space of all possible quantum states, called *Hilbert space*, is infinite-dimensional for the position of a single particle. That's why qubits are so much easier to think about. Two dimensions are easier to visualize than infinite dimensions.

When there are only two components in our quantum state, as opposed to infinitely many, it can be hard to think of the state as a "wave function." It's not very wavy, and it doesn't look like a smooth function of space. The right way to think about it is actually the other way around. The quantum state is not a function of ordinary space, it's a function of the abstract "space of measurement outcomes," which for a qubit only includes two possibilities. When the thing we observe is the location of a single particle, the quantum state assigns an amplitude to every possible location, which looks just like a wave in ordinary space. That's the unusual case, however; the wave function is something more abstract, and when more than one particle is involved, it becomes hard to visualize. But we're stuck with the "wave function" terminology. Qubits are great because at least the wave function has only two components.

This may seem like an unnecessary mathematical detour, but there are immediate payoffs to thinking about wave functions as vectors. One is explaining the Born rule, which says that the probability for any particular measurement outcome is given by its amplitude squared. We'll dive into details later, but it's easy to see why the idea makes sense. As a vector, the wave function has a length. You might expect that the length could shrink or grow over time, but it doesn't; according to Schrödinger's equation, the wave function just changes its "direction" while maintaining a constant length. And we can compute that length using Pythagoras's theorem from high-school geometry.